

Categorize-then-Predict

Trustworthy PM2.5 where there is no sensor — and a few days ahead

Kampala, Uganda · 39 low-cost sensors · free data only

The question Can a categorize-then-predict pipeline give an urban planner trustworthy PM2.5 where there is no sensor, and a few days ahead, without collapsing? This deck reports the honest answer — out-of-site and out-of-time, negative results included.

From 60+ Papers Down to 4

60+ screened Air-quality + ML + remote-sensing literature for low-resource cities.

4 anchors kept Each builds on the last — precedent, levers, honest evaluation.

1 gap Models look great on a random split, then collapse at a new place.

→ The gap — generalization to unseen places — is exactly what the 4th cap attacks.

Each Paper Builds on the Last

Adong 2025 Kampala PM2.5 from satellite AOD alone — weather deliberately excluded. Our precedent.

Weather lever ERA5 weather is the cheap, generalizable signal the precedent left out.

Honest evaluation African spatial-CV ~ 0.13 vs random > 0.90 — collapse is the norm, not the exception.

Land-use regression Road density / distance-to-road / terrain transfer to new locations.

→ Together: cheap data can work — but only if evaluated honestly and given place-structure.

From “Which Lever?” to “Categorize-then-Predict”

Old framing (3rd cap) “Which cheap data lever helps estimate PM2.5?” → a finding (weather helps, LST is inert), but the tool collapses at a sensor-less place.

New framing (4th cap) A clinic, not one doctor: a triage nurse sorts each place into a kind, then the specialist for that kind predicts.

Two metrics — kept separate end to end:

NOW • nowcast PM2.5 today at a sensor-less place. Honest test = leave-sites-out. Bar = the blind pooled model.

FUTURE • forecast PM2.5 +1...7 days at a known place. Honest test = chronological. Bar = persistence (“same as yesterday”).

→ Different question, different honest test, different bar, different best model.

39 Sensors • One Merged Table • Free Inputs Only

39

AirQo sensors

13,944

site-days

~84%

daily coverage

38

$\mu\text{g}/\text{m}^3$ mean PM2.5

Predictors (free, available anywhere) ERA5 weather (temp • dew-point • wind • pressure • rain) + location + calendar. LST carried but proven inert.

Banned as predictors pm10 (same instrument as the target, corr $\approx$ 0.9) and n_obs (a by-product of the daily mean) — neither exists at a sensor-less place, so using them would manufacture a fake win.

→ Honest features first: nothing a planner could not actually obtain at a new address.

Looks Great — Until You Point It Somewhere New

0.81

leaky random-split R^2

0.31

honest new-site R^2

~0

one new-site fold

Why it matters The headline $R^2 \approx 0.81$ is a leaky random split — it tests on the same sensors it trained on (gap-filling). Graded honestly (hide whole sensors, or forecast the real future) it collapses, and a “same-as-yesterday” guess beats it out-of-time. A planner needs PM2.5 where there is no sensor.

→ The whole 4th cap exists to stop this collapse — without faking the test.

39 Sensors → 3 Behavioural Kinds of Place

24

low-pollution / well-ventilated — the city baseline

PM2.5 ≈ 31.7

12

high-pollution / well-ventilated — hotter spots, still windy

PM2.5 ≈ 47.5

3

moderate / stagnant — low wind + humid, a robust micro-regime

PM2.5 ≈ 46.5

Robust + portable Same 3 stagnant sites at every k; k-means agrees with Ward. Categories are spatially intermixed — they capture behaviour, not geography. A new site is categorized from its signature, snapped to the nearest centroid.

→ Kampala is homogeneous in level; the structure is low-vs-high pollution + 3 stagnant outliers.

Knowing the Category Beats the Blind Model

New-site R^2 (leave-sites-out) — climb only if the rung below holds:

blind (no category)

0.316

+ category · k=4 (tuned)

0.558

+ category + geodata · k=4

0.564

Significant Tuning categories to k=4 (by honest R^2 , not silhouette) lifts the deploy-only nowcast $0.48 \rightarrow 0.56$. Paired bootstrap $+1.07 \mu\text{g}/\text{m}^3$, CI $[+0.33, +1.90]$.

Soft routing wins Category-as-a-feature tolerates a wrong category; mixture-of-experts hard-routes and collapses ($0.50 \rightarrow 0.30$) under realistic cold-start.

→ Ceiling: leave a whole category out → negative. Interpolates among known kinds, can't extrapolate.

The Model Is Not the Bottleneck

New-site R^2 · same +category features · four learners:

0.529

RandomForest

0.482

XGBoost

0.469

LightGBM

0.465

CatBoost

Best: RandomForest Lowest leaky score yet highest new-site score — it overfits least, transfers best. But all 4 cluster (0.47–0.53).

Conformal — fixed Calibrating on held-out SITES (group conformal) lifts new-site 90%-interval coverage 74.8% → 85.8%. Bands widen honestly.

→ The category feature (+0.16) moved R^2 far more than any model swap. Features > learner.

Free Map Data Nowcasts a Sensor-less Site

0.32

blind

0.42

+ geodata · ZERO sensors

0.55

+ category + geodata

Land-use regression, sourced free OSM road density + distance-to-nearest-major-road (Overpass, 43,870 ways) + SRTM elevation. Static, need ZERO pollution observations. dist-to-major-road is the clearest signal (closer → higher PM2.5).

A complement, not a substitute Geodata can't name the category (54% < 62% baseline) — it adds independent signal. Combined is best, and most stable (± 0.046).

→ The bridge from “a finding” to “a tool a planner can point at any address.”

A Real Forecaster Holds $R^2 \approx 0.43$ a Week Out

R^2 by horizon (chronological, out-of-time):

horizon	persistence	GBM-on-lags	LSTM
+1 day	0.34	0.43	0.51
+3 days	0.11	0.37	0.48
+7 days	0.02	0.33	0.43

LSTM earns its weight Global LSTM + category embedding beats the GBMs at every horizon (+0.04...+0.12), lead widening with horizon. All beat persistence. Correction: forecasting does NOT decay to ~0 by +7d.

→ Persistence-beats-model was true only for the nowcast model — not a purpose-built forecaster.

From a Number to a Safety Status

EPA PM2.5 safety bands — what a city official actually needs:

Elevated

$\leq 35.4 \mu\text{g}/\text{m}^3$

High

35.5 – 55.4

Dangerous

> 55.4

Graded by the BAND it lands in — same honest tests:

New-site (no sensor) Exact tier 68% • within one tier 98.5% • vs 52% majority baseline. A 2-tier error (a Dangerous area called safe) for only 3.6% of days.

Forecast +1→+7d Exact tier 70% → 68% • within one tier ~98% • beats the persistence-band baseline at every horizon.

→ R^2 punished 47-vs-52 misses; in safety tiers those are the SAME band — so the tool is deployable.

Catching the Days That Matter — a Planner's Dial

Lower the Dangerous alert cutoff → catch more dangerous days, at a chosen false-alarm cost (honest new-site):

alert if predicted >	dangerous-day recall	false-alarm rate
55.4 (strict EPA)	0.51	4%
50	0.67	8%
45 (suggested)	0.80	17%
40	0.91	30%

Why it works Dangerous days are a humid, stagnant weather regime (humidity + dew-point separate them most) — the model keys on weather to flag them.

Honest negative A dedicated class-weighted detector did NOT beat simply lowering the regressor's threshold. The simpler lever wins.

→ The planner owns the operating point: 80% of dangerous days caught at a 45 $\mu\text{g}/\text{m}^3$ alert cutoff.

The Reframe Works — With Honest Limits

What worked Category lifts new-site R^2 0.32→0.48 (significant). Free geodata: 0.42 zero-deploy, 0.55 combined. LSTM holds 0.43 at +7d.

What didn't Mixture-of-experts collapses under cold-start. Category×month prior hurts under noise. Geodata can't name the category.

Bounded Leave-a-whole-category-out goes negative — can interpolate among known kinds, can't extrapolate to an unseen regime.

Two prior claims corrected (verified) 1) Forecasting does not decay to ~0 by +7d — an artefact of using a nowcast model to forecast. 2) “Categorization can't rescue a homogeneous city” is too pessimistic — it lifts new-site R^2 by +0.16; true only for unseen regimes.

→ Honesty over headlines — every win holds out-of-site and out-of-time, or it isn't reported.

A Cheap, Honest PM2.5 Safety Tool for a Low-Resource City

The contribution Free data + behavioural categories + land-use geodata turn a collapsing model into a deployable safety-triage tool: classify a never-instrumented area's safety tier (68% exact / 98.5% within one tier) and forecast a week ahead — evaluated the way a planner will actually use it.

Done • safety tiers band evaluation + planner-tunable danger alerting (80% recall)

Done • sharper model $k=4$ categories (0.56) + group conformal (86% coverage)

Next • more places 39 sites is small — more sensors/categories sharpen routing

Next • push recall cost curves to 90% danger recall • tighter conformal to 90%

→ Reproducible: fixed seeds • honest splits • scripts + result CSVs in 4th-cap-handover/analysis/.